9 James and Peter cycled along the same 50 km route.

James took $2\frac{1}{2}$ hours to cycle the 50 km.

Peter started to cycle 5 minutes after James started to cycle.
Peter caught up with James when they had both cycled 15 km.

James and Peter both cycled at constant speeds.

Work out Peter's speed.

James: 2½ hours = 150 minutes.

	Min	Km	
	150	50	
÷ 50			÷ 50
	3	1	
× 15			× 15
	45	15	

1 mark
Find his rate: 1 km in 3 minutes

1 mark
James takes 45 minutes for 15 km

Peter starts 5 minutes later: 45 – 5 = 40 minutes.

1 mark
Subtract the delayed start

	Min	Km	
	40	15	
$\times \frac{3}{2}$			$\times \frac{3}{2}$
	60	22.5	

In 60 minutes Peter cycles 22.5 km.

1 mark
Scale the distance to one hour

1 mark
Peter's speed is 22.5 km/h

22.5 km/h

(Total for Question 9 is 5 marks)